Hugo Marquerie 26/10/2025

Prop proyeccion ortogonal convexo cerrado

Proposición 1. *Sea H un espacio de Hilbert y $S \subset H$ no vacío, cerrado y convexo*

$$\implies \forall x \in H : \exists! y \in S : \|x - y\| = \min\{\|x - z\| : z \in S\}.$$

Denotamos $y = P_S(x)$ y la aplicación $P_S \colon H \to S$ se llama ***proyección ortogonal****.*

***Demostración*:** Repetir la prueba del `teo-cerrado-convexo-hilbert-imp-exists-min-norma/`Teore trasladando el origen a x. ■

Referenciado en

- `teo-carac-proyeccion-ortogonal-subespacio-cerrado`
- `prop-carac-proyeccion-ortogonal-convexo-cerrado`
- `teo-con-fundamental-iff-sistema-ortogonal-completo`
- `teo-proyeccion-ortogonal-sistema-ortonormal-formula`

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